



Storm Sequence Generator — Test Results

Automated Test Documentation for Model Governance

MKM Research Labs

Version 1.0 — 04-April-2026

David K Kelly

CONFIDENTIAL — PROPRIETARY

Legal Notice

PROPRIETARY AND CONFIDENTIAL

This document, including all algorithms, models, methodology, formulas, calculations, and intellectual concepts contained herein, constitutes the exclusive intellectual property and confidential trade secrets of MKM Research Labs (“MKM”).

Any usage, reproduction, distribution, modification, or disclosure of this document or any portion thereof, without the express prior written authorisation from MKM Research Labs is strictly prohibited and constitutes an infringement of intellectual property rights.

The document is provided on an “as is” basis. MKM Research Labs makes no representations or warranties of any kind, express or implied, regarding its accuracy, reliability, or suitability for any particular purpose.

All rights reserved. © 2021–2026 MKM Research Labs.

Governed by the laws of the United Kingdom.

1 Test Results — Storm Sequence Generator (MKM-SS-001)

Tests run on **04 April 2026 at 14:48:33**.

Table 1: Test Results Summary — Storm Sequence Generator (MKM-SS-001)

Total Tests	17
Passed	17
Failed	0
Skipped	0
Duration	0.00s

Table 2: Detailed Test Results — Storm Sequence Generator

Test Description	Acceptance Criteria	Result	Time
Duration must not exceed MAX_PRECIP_END_HOUR (156).	<code>result.storms[0].duration_hours <= MAX_PRECIP_END_HOUR</code>	Pass	0.000s
Isolated has one storm	<code>isinstance(result, StormSequence);</code> <code>result.num.storms == 1 (+2 more)</code>	Pass	0.000s
Aggregate stats	<code>result.max.intensity_factor >=</code> <code>result.avg.intensity_factor;</code> <code>result.total.precipitation_mm > 0</code> (+1 more)	Pass	0.000s
Catchment id in storms	<code>result.storms[0].catchment_id ==</code> "rhine"	Pass	0.000s
Cluster sequence	<code>result.sequence_type in</code> (SequenceType.CLUSTER.value, Sequ...	Pass	0.000s
force_sequence_type='doublet' → always doublet.	<code>result.sequence_type ==</code> SequenceType.DOUBLET.value; <code>result.num.storms == 2</code>	Pass	0.000s
force_sequence_type='isolated' → always isolated.	<code>result.sequence_type ==</code> SequenceType.ISOLATED.value; <code>result.num.storms == 1</code>	Pass	0.000s
Minimal intensity (5% sequence probability) mostly produces isolated.	<code>isolated >= 10</code>	Pass	0.001s
Generate multiple sequences without error.	<code>isinstance(result, StormSequence);</code> <code>result.num.storms >= 1</code>	Pass	0.000s
Persistent sequence	<code>result.sequence_type in</code> (SequenceType.PERSISTENT.value, S...	Pass	0.000s
Returns storm sequence	<code>isinstance(result, StormSequence)</code>	Pass	0.000s

Test Description	Acceptance Criteria	Result	Time
All sequences must have to- tal_duration != 168h.	result.total_duration_hours <= 168.0	Pass	0.000s
Storms have correct fields	storm.duration_hours > 0; storm.intensity_factor > 0 (+2 more)	Pass	0.000s
Custom catchment	gen.catchment_id == "rhine"; gen.base_precipitation == CATCHMENT_BASE_PRECIP["rhine"]	Pass	0.000s
Default catchment	gen.catchment_id == "thames"; gen.base_precipitation == CATCHMENT_BASE_PRECIP["thames"]	Pass	0.000s
Same seed → same sequence.	seq1.num_storms == seq2.num_storms; seq1.sequence_type == seq2.sequence_type	Pass	0.000s
Unknown catchment defaults to 35	gen.base_precipitation == 35.0	Pass	0.000s

1.1 Test Document History

Date	Version	Description	Author
12-Mar-2026	1.0	Initial test suite (116 tests)	D. Kelly
27-Mar-2026	1.1	64 test(s) removed (total: 52)	D. Kelly
04-Apr-2026	1.2	35 test(s) removed (total: 17)	D. Kelly